14. (2.75 points)

Given the following output from a generalized linear model fitted to a triangle of loss development data:

Standardized Pearson Residuals

Accident Year	12 Months	24 Months	36 Months	48 Months	60 Months	72 Months
2010	1.68	2.65	-4.70	-2.78	3.37	0.00
2011	7.74	-1.34	-7.33	-1.91	-3.61	
2012	5.72	1.94	-6.91	4.98		
2013	-1.22	0.12	-1.94			
2014	1.67	-1.87				
2015	0.00					

Standard Deviations of Standardized Pearson Residuals

Accident Year	Standard Deviation	Accident Year Range	Standard Deviation
2010	3.571	2010 to 2011	4.463
2011	5.563	2012 to 2013	4.345
2012	5.797	2014 to 2015	2.503
2013	1.045	2010 to 2012	4.741
2014	2.503	2013 to 2015	1.537

Fitted Cumulative Losses

Accident Year	12 Months	24 Months
2012	3,500	7,612
2013	4,210	8,749
2014	5,400	10,654

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(14 continued)

An actuary reviewing the output notices heteroscedasticity in the residuals. The actuary decides to adjust for this by calculating variance parameters before running the sampling algorithm.

Given the following residual index values from one iteration of the sampling algorithm:

Accident Year	Sample Residual Index (row, column)
2012	5, 1
2013	1, 3
2014	4, 3

a. (2.25 points)

Calculate the actuary's sampled incremental losses for accident years 2012-2014 between 12 and 24 months for the sample under consideration.

b. (0.5 point)

Explain why heteroscedastic residuals might cause issues when using a bootstrapping technique to estimate variance of unpaid claim estimates.

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QUESTION 14

TOTAL POINT VALUE: 2.75	**LEARNING OBJECTIVE:** A08: Identify data issues and related model adjustments for reserving models. Also A10: Develop a distribution of reserves using weights and multiple stochastic models.

SAMPLE ANSWERS

Part a: 2.25 points

Sample Answer 1

* Need to adjust residuals for heteroscadasticity.

* Group residuals by AY:

2010 – 2012 have similar standard deviation

2013 – 2015 have similar standard deviation

$\sigma_{2010-2012} = 4.741$

$\sigma_{2013-2015} = 1.537$

2013 – 2015 residual adj = 4.741 / 1.537 = 3.085

2010 – 2012 residual adj = 4.741 / 4.742 = 1.000

$\max \sigma = 4.741$

$q^*(w,d) = \left(r^*\sqrt{m_{w,d}} + m_{w,d}\right)$

$q^*(3,2) = \left(5.151\sqrt{4{,}112} + 4{,}112\right)$

$q^*(4,2) = \left(-1.524\sqrt{4{,}539} + 4{,}539\right)$

$q^*(5,2) = \left(-1.94\sqrt{5{,}254} + 5{,}254\right)$

$m_{3,2} = 7{,}612 - 3{,}500$

$m_{4,2} = 8{,}749 - 4{,}210$

$m_{5,2} = 10{,}654 - 5{,}400$

AY	Unadjusted Sample Residual	AY Sampled From	Hetero-Adjustment	Adjusted r^*	$q^*(w,d)$
2012	-1.67	2014	3.085	5.152	4,442.37
2013	-4.70	2010	1 / 3.085	-1.524	4,436.32
2014	-1.94	2013	1.000	-1.94	5,113.38

Sample Answer 2

	Sample Residual	SD(Residual)	SD(AY Group)
2012	1.67	1.537	4.741
2013	-4.7	4.741	1.537
2014	-1.94	1.537	1.537

Adjust SD based on groupings

Group 2011 – 2012

Group 2013 – 2015

Sample hetero adjusted residual

2012 $1.67 \times \frac{4.741}{1.537} = 5.15$

2013 $-4.7 \times \frac{1.537}{4.741} = -1.524$

2014 $-1.94 \times \frac{1.537}{1.537} = -1.94$

	Fitted Incremental Losses	Resampled Incremental Losses
2012	4,112	$5.15 \times \sqrt{4{,}112} + 4{,}112 = 4{,}442$
2013	4,539	$-1.52 \times \sqrt{4{,}539} + 4{,}539 = 4{,}436$
2014	5,254	$-1.94 \times \sqrt{5{,}254} + 5{,}254 = 5{,}113$

Sample Answer 3

Accident years 2010 to 2012 have similar standard deviations, so group together. Other accident years do not have similar standard deviation. Selecting groups as follows:

AY	Std Dev	Hetero Group	Adj Factor
2010	3.571	1	4.471/4.471 = 1.00
2011	5.563	1	1.00
2012	5.797	1	1.00
2013	1.045	2	4.471/1.045 = 4.537
2014	2.503	3	4.471/2.503 = 1.894

AY	Adj Residual (r*)	Adj Residual (r*/h)
2012	1.67 x 1.894 = 3.163	3.163 / 1.00 = 3.163
2013	-4.70 x 1.000 = -4.700	-4.70 / 4.537 = -1.036
2014	-1.94 x 4.537 = - 8.802	-8.802 / 1.894 = -4.648

AY	Sampled Loss
2012	$(7{,}612 – 3{,}500) + 3.163(7{,}612 – 3{,}500)^{1/2} = 4{,}315$
2013	$(8{,}749 – 4{,}210) – 1.036(8{,}749 – 4{,}210)^{1/2} = 4{,}469$
2014	$(10{,}654 – 5{,}400) – 4.648(10{,}654 – 5{,}400)^{1/2} = 4{,}917$

Part b: 0.5 point

Sample Answer 1

The bootstrapping process assumes we can sample residuals from anywhere in the triangle. If the variance of residuals differs then our assumption of independent residuals is

not valid. Adjusting for this keeps us from having overstated or understated estimated incremental losses during each iteration (depending on how the variance in that cell relates to other cells) and keeps the bootstrap variance of loss estimate from being artificially distorted.

Sample Answer 2

Bootstrapping technique requires the residuals to be IID, then we can sample residuals from any place of the triangle. Heteroscedastic residuals violates this requirement.

EXAMINER'S REPORT

Very few candidates earned full credit for both parts of this question.

Part a

- Candidates were expected to know how to select hetero-groups based on the standard deviations provided. Solutions with different hetero-groups were given credit as long as the candidate provided justification.
- Candidates needed to recognize that sampled residuals should be adjusted for heteroscedasticity.
- Candidates had difficulty earning full credit for this part.
- Common errors include the following:
  - No justification provided for selected hetero-groups.
  - Hetero-adjustment factors were calculated using the wrong standard deviations
  - Entirely skipped hetero-adjustments.
  - Applied hetero-adjustment factors to the wrong sampled residuals.
  - Candidates failed to divide by the hetero-adjustment factors to bring the adjusted residuals back to the original distribution.
  - Some candidates did not use incremental fitted losses and applied the process to cumulative losses.

Part b

- Candidates were expected to know that residuals are independent and identically distributed in order to sample residuals from the whole triangle.
- Generally, candidates did not earn full credit for this part.
- Common errors include the following:
  - Noting that bootstrapping requires residuals that are independent and identically distributed, but did not fully explain the issue when the assumption is violated.
  - Other candidates used terms such as "the variance", "variance of incremental losses", or "losses are iid" without mentioning that it's the <u>residuals</u> that have to be IID.
  - Many candidates mentioned that it would "increase the variance of the unpaid claim estimate" which is not always the case.
  - Full or partial credit was given to candidates who explained the variance of the unpaid claims would increase or decrease due to heteroscedastic residuals.